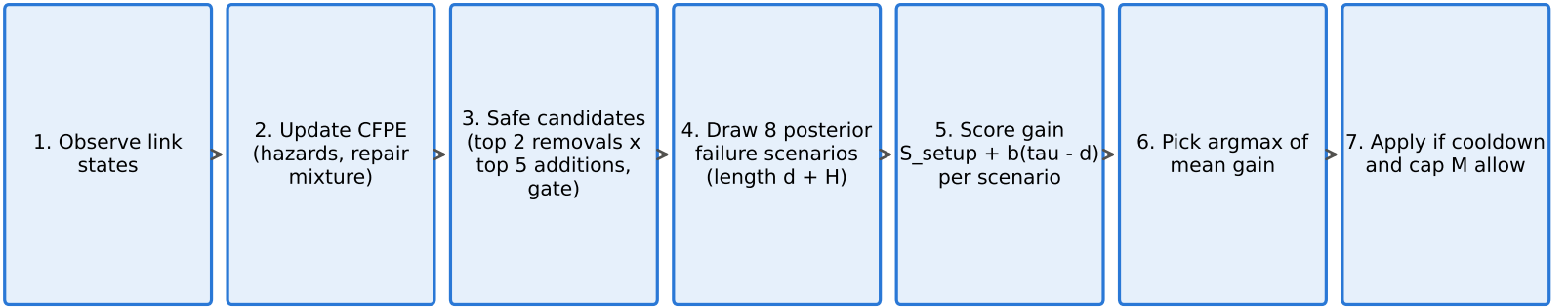

Not in the final algorithm: rollout shadow price for timing; full-horizon rerank (exploratory variants only).